Jinrui (Jerry) Gou

Computer Science and Engineering
Tandon School of Engineering
New York University

Tel : (347) 200 5159
Email: jg6226@nyu.edu
HomePage: *j9rrygou.github.io*

RESEARCH INTERESTS

Information Retrieval, Efficient Search and Ranking, Nearest Neighbor Search, Recommender Systems

EDUCATION

New York University — Brooklyn, NY
Ph.D. in Computer Science — 09/2021-Present
Advisor: Professor Torsten Suel
Cumulative GPA: 3.97/4.0

M.S. in Computer Science — 09/2019-05/2021
Cumulative GPA: 4.0/4.0

Sun Yat-Sen University — Guangzhou, China
B.E. in Computer Science & Technology — 09/2015-06/2019
Cumulative GPA: 90/100 (3.8/4.0)
Mathematics Courses GPA: 95/100

KNOWLEDGE

- **Languages:** C/C++, Python, SQL, AMPL/CPLEX, Latex, MATLAB, Bash, HTML5.
- **Tools:** Scikit-learn, PyTorch, TensorFlow, Transformers, XGBoost, Numpy, Matplotlib, CUDA.
- **Courses Enrolled:** Computer Architecture, Natural Language Processing, Machine Learning (Rank 1^{st} in Class of 43 students), Deep Learning, Web Search Engines, Design & Analysis of Algorithms (Rank 1^{st} in Class of 47 students), Network Design and Algorithms (Rank 2^{nd} in Class of 20 students), Algorithmic Machine Learning and Data Science.

RESEARCH EXPERIENCE

Ph.D. Research Assistant for Prof. Torsten Suel, NYU — 09/2021-Present

1. Scalable Candidate Generation and Threshold Estimation for Fast Top-K Information Retrieval
   - Develop efficient early termination algorithms for ranking scoring functions, enabling fast top-k retrieval over large-scale document collections.
   - Design specialized prefix index structures and fine-tune storage layout for cache-efficient access during ranking.
   - Perform fast candidate lookups using highly optimized index structures to improve ranking quality and recall.
   - Implement end-to-end retrieval and ranking pipeline in C++ evaluated across multiple corpora and ranking models.
   - Achieve 5–10$\times$ speedup with negligible end-to-end quality loss compared to optimized baseline algorithms, with direct applicability to recommendation ranking pipelines.
2. Graph-Structured Models for Approximate Nearest Neighbor Search and Scalable Retrieval

- Study the theoretical limits of constructing sparse navigable graphs for nearest neighbor search in high-dimensional embedding spaces, directly applicable to item retrieval in large-scale recommender systems.
- Design efficient randomized and deterministic graph construction algorithms achieving $O(\sqrt{n}\log(n))$ average degree for any point set and distance function.
- Bound the maximum degree and average degree of navigable graphs using mathematical proof.
- Prove nearly matching lower bounds showing $\Omega(n^{1/2})$ average degree is necessary for navigable graphs on random point sets.
- Develop efficient heuristic graph pruning algorithms to drastically reduce the average degree of a graph while maintaining navigability.
- Propose Distance Adaptive Beam Search, with a smarter termination condition that replaces the standard fixed beam width, improving search efficiency on graph-structured indexes.
- Prove guaranteed search accuracy on navigable graphs and achieve 30–40% fewer distance computations over state-of-the-art graph indexes (HNSW, Vamana).

3. Sparse Learned Index Structures Optimization with Large Language Models
   - Contribute to a project combining LLM-based document expansion (Llama 2) with Transformer-based term importance prediction, enhanced by hard negatives and distillation.
   - Assist in dataset preparation and ensure experiment reproducibility across benchmark collections.
   - Conduct evaluation experiments using PISA search engine toolkit and contribute to project presentation materials.

Graduate Research Assistant for Prof. Yong Liu, Networked Systems Lab, NYU 09/2019-05/2021

1. Realtime Mobile Network Prediction for High QoS
   - Collect mobile 4G/LTE bandwidth traces in metro area of NYC.
   - Develop LSTM RNN models for high-accuracy realtime bandwidth prediction.
   - Develop Gradient Boosting models for handoffs predictions between 4G and 5G access modes.
2. Joint Traffic Routing and Computation Server Placement in Edge Cloud Networks
   - Build quasi-convex optimization model to balance traffic and computation resource allocation.
   - Use AMPL/CPLEX and Python optimization packages to obtain optimal feasible solutions.

PUBLICATIONS

1. Yousef Al-Jazzazi*, Haya Diwan*, **Jinrui Gou***, Cameron Musco*, Christopher Musco*, Torsten Suel*, "Distance Adaptive Beam Search for Provably Accurate Graph-Based Nearest Neighbor Search", *NeurIPS 2025: Advances in Neural Information Processing Systems, December 2025*[1]
2. **Jinrui Gou**, Antonio Mallia, Yifan Liu, Minghao Shao, and Torsten Suel, "Fast and Effective Early Termination for Simple Ranking Functions", *ACM SIGIR Conference on Research and Development in Information Retrieval, July 2025.*

[1] All authors contributed equally.

3. Haya Diwan*, **Jinrui Gou***, Cameron Musco*, Christopher Musco*, and Torsten Suel*, "Navigable Graphs for High-Dimensional Nearest Neighbor Search: Constructions and Limits", *NeurIPS 2024: Conference on Neural Information Processing Systems, December 2024.*[2]
4. **Jinrui Gou**, Yifan Liu, Minghao Shao, and Torsten Suel, "Beyond Quantile Methods: Improved Top-K Threshold Estimation for Traditional and Learned Sparse Indexes", *IEEE International Conference on Big Data, December 2024.*
5. Soyuj Basnet, **Jinrui Gou**, Antonio Mallia, and Torsten Suel, "DeeperImpact: Optimizing Sparse Learned Index Structures", *in ReNeuIR at SIGIR 2024: The Third Workshop on Reaching Efficiency in Neural Information Retrieval, July 2024.*
6. Lifan Mei, **Jinrui Gou**, Jingrui Yang, Yujin Cai, and Yong Liu, "On Routing Optimization in Networks with Embedded Computational Services", *IEEE Transactions on Network and Service Management, September 2024.*
7. Lifan Mei, **Jinrui Gou**, Yujin Cai, Houwei Cao, and Yong Liu, "Realtime mobile bandwidth and handoff predictions in 4G/5G networks", *Computer Networks, February 2022*.

EMPLOYMENT EXPERIENCE

Applied Scientist Intern, Amazon, Sunnyvale, CA Summer 2026

- Work on applied machine learning and large-scale search systems.
- Explore efficient model serving and retrieval components under practical latency constraints.

Quantitative Research Intern, Cantor Fitzgerald, New York Summer 2025

- Develop short-term stock price movement prediction models (10s, 20s, and 30s horizons) using time series analysis and ensemble methods including Random Forest and customized XGBoost models.
- Utilize kdb+/q for scalable preprocessing of high-frequency time series market data, performing feature engineering including joins, alignment, and technical indicators (e.g., MACD, KDJ).
- Incorporate proprietary dark pool trading data as additional predictive features, building end-to-end ML pipelines for model training and evaluation to enhance prediction accuracy.

Course Assistant, Web Search Engines & Principles of Database Systems, NYU Spring/Fall 2025

- Design homework problems, conduct demo sessions, and grade students' assignments and exams.
- Assist the professor by answering students' questions on Piazza and holding weekly TA office hours.

Teaching Assistant, ECE-GY 9343 Data Structures and Algorithms, NYU Spring–Fall 2020

Design homework problems, grade assignments and exams, and hold weekly office hours.

REFERENCES

Prof. Torsten Suel	**Prof. Christopher Musco**	**Prof. Yong Liu**
New York University	*New York University*	*New York University*
torsten.suel@nyu.edu	*cmusco@nyu.edu*	*yongliu@nyu.edu*

[2] All authors contributed equally.